Snowball RL: controlled probability check

Fresh checkpoint restores · identical tokens · zero optimizer updates

Both primary comparisons completed on the same 56 sequences (84,274 response tokens), with zero optimizer updates and verified matching learner/serving parameters. Mean absolute log-probability disagreement rose from 0.05719 at checkpoint 384 to 0.17820 at checkpoint 512 (3.12-fold). The fraction of token probabilities differing by more than 2× rose from 1.827% to 7.641%. Aggregating the paired archive texts for each question, disagreement increased on all 28 questions and all 14 categories.

Forcing the learner to use vLLM's response expert choices reduced mean disagreement by 65.4% at checkpoint 384 and 70.8% at checkpoint 512. Large disagreements fell to 0.234% and 1.683%, respectively. The intervention reduced the mean gap on all 28 questions at each checkpoint, and recorded expert-set agreement under replay was 100%. Native expert-set disagreement increased more modestly, from 17.55% to 19.24% of measured token/layer positions. The much larger probability effect is consistent with increased sensitivity to routing differences; the experiment does not identify its first numerical trigger.

The checkpoint-by-text comparison shows a strong interaction. On archive-384 text, the gap rose from 0.04069 to 0.24132. On archive-512 text, it instead fell from 0.08514 to 0.07131. Thus the aggregate increase is driven by the later checkpoint becoming especially inconsistent on earlier response traces. A claim that agreement worsened uniformly on all text would be incorrect. The fixed-token experiment demonstrates a persistent checkpoint-dependent effect, while also showing that response content substantially changes its size.

Checkpoint / mode	Mean |δ|	Mean k3	Beyond 2×	ESS / N
384: native	0.05719	0.37549	1.827%	0.0006146
384: forced response routes	0.019788	0.0037775	0.234%	0.9881
512: native	0.1782	2128.4	7.641%	2.366e-05
512: forced response routes	0.052085	0.034637	1.683%	0.279

The main comparisons are mean |δ| and the fraction beyond 2×. On archived text, very low reference probabilities can produce extreme ratios; k3 and ESS can consequently be dominated by a few tokens. These fixed-panel k3 values are not estimates of current-policy KL and must not be compared numerically with the training-time k3 curve.

Controls

Checkpoint 384: all 16 serving workers and 32 learner ranks passed verification. 314,452,475,904 serving bytes and 183,590,191,104 learner-peer bytes were compared, with zero mismatches. All 832 full-FP32 router weight/bias comparisons agreed exactly. Mean |δ| for repeated vLLM scoring: 0.016566; rebatched learner versus original learner: 0.018935.

Checkpoint 512: all 16 serving workers and 32 learner ranks passed verification. 314,452,475,904 serving bytes and 183,590,191,104 learner-peer bytes were compared, with zero mismatches. All 832 full-FP32 router weight/bias comparisons agreed exactly. Mean |δ| for repeated vLLM scoring: 0.050346; rebatched learner versus original learner: 0.048795.

This establishes a causal contribution of response-route selection to the probability mismatch in this diagnostic. It does not establish that the mismatch caused the task-accuracy regression; that requires a controlled training comparison. Fresh restores do not reconstruct the original in-memory serving state. These jobs used GB200 hardware and new physical placements, whereas the training run used H100s.

The panel is small and family-balanced, with responses capped at 2,048 tokens. Prompt routes remain native during replay. The residual gap is comparable to repeat/rebatch variation within a backend, but is not zero and can still include prompt-routing effects. The optional decode control at checkpoint 384 failed before generation, after its complete primary comparison had been saved; the checkpoint-512 control completed.

Decode and repeatability controls

These measurements use fresh checkpoint-specific responses, separate from the 56-sequence fixed panel. All ratios in the table below use the probability recorded during vLLM autoregressive decoding as the denominator. Prefill rescoring checks numerical differences within vLLM itself.

Checkpoint 384: no auxiliary decode measurements. Its primary fixed-panel comparison completed and was archived before the auxiliary request failed.

Checkpoint / comparison	Mean |δ|	Mean k3	Beyond 2×	ESS / N
512: vLLM prefill / decode	0.052419	0.028892	1.938%	0.9629
512: native learner / decode	0.063647	0.037967	2.166%	0.9581
512: forced routes / decode	0.022577	0.0077287	0.823%	0.988

At checkpoint 512, 32 generated responses to four questions supplied 7,896 scored tokens and 30 distinct response-token sequences. All requests used the shared seed 20261017; repeated prompts are not treated as independent samples. This control uses the probabilities and expert routes recorded during actual autoregressive decoding.

Relative to the decode probabilities, the learner’s mean absolute log-probability gap fell from 0.06365 to 0.02258 under response-route replay (a 64.5% reduction). The fraction of probability ratios outside ½–2 fell from 2.166% to 0.823%; k3 fell from 0.03797 to 0.00773. This supports the routing mechanism on actual generated tokens, beyond the archived-text prefill comparison.

vLLM prefill rescoring itself differed from vLLM decoding: mean absolute log-probability difference 0.05242, with 1.938% of ratios outside ½–2. Execution-mode sensitivity therefore also matters. The fixed-panel k3 values are especially tail-sensitive: one token contributes 56.7% of checkpoint 512’s native k3, and one token contributes 98.8% of the separate fresh-restore vLLM control’s k3. These descriptive statistics must not be interpreted as current-policy KL or as task accuracy.

This control covers four questions and short response prefixes. It cannot establish the magnitude of decode/prefill differences over complete, long reasoning traces. It does not provide a controlled checkpoint comparison with fixed response tokens.

Separate fresh restores

An additional checkpoint-384 control compares vLLM scores saved in two separate fresh restores, both passing weight verification. The first attempt committed its vLLM scores before failing to archive the learner scores. On identical tokens, the two vLLM runs have mean |δ| 0.035545, k3 5.1702, 1.060% of ratios beyond 2×, and 90.232% expert-set agreement.

Methods and definitions

The fixed panel contains 28 randomly selected validation questions, two in each of 14 available families (seed 20261017). Each question contributes its archive-384 and archive-512 response: 56 sequences and 84,274 scored response tokens. Original prompt token IDs are retained. Responses are limited to their first 2,048 tokens, or their full length when shorter. Archive 384 contributes 52,985 tokens and archive 512 contributes 31,289. This is a family-balanced diagnostic panel, not an accuracy evaluation or a representative sample of training tokens. The source archives report model versions 383 and 511, respectively, because evaluation precedes the next update. The checkpoints used to rescore both sources are completed updates 384 and 512. Historical archive probabilities are not used.

Each checkpoint is restored into fresh MarinSkyRL workers using 48 CoreWeave GB200 GPUs: 32 learner GPUs with tensor parallelism 1, pipeline parallelism 2 and expert parallelism 16; two rollout replicas with data/expert parallelism 8 each. The original training used H100 GPUs. The learner and rollout engines retain their native implementations. vLLM uses FlashAttention 4 and the Triton MoE backend in both jobs. Model weights are BF16; router state is checked separately in FP32. Native broadcast installs serving weights. An independent expert-block scratch replay then verifies the installed bytes without reinstalling them, followed by full-FP32 readback of all router weights and biases.

vLLM scores the fixed text using prompt log probabilities, plus a final-token candidate score. Temperature is 1, top-p is 1, top-k is disabled, and prefix caching is disabled. Its expert choices are captured on that same forward. Megatron then scores the exact tokens in three modes: native (microbatch 1), reordered/rebatched (microbatch 2), and forced response-router replay. Replay sets the response input-token expert choices to vLLM's choices; prompt routes and expert combine weights remain native. The final response token has no captured input route because it was not fed back to the model. vLLM scoring is repeated after the learner forwards. No optimizer update occurs; the training-update method is guarded to raise if called.

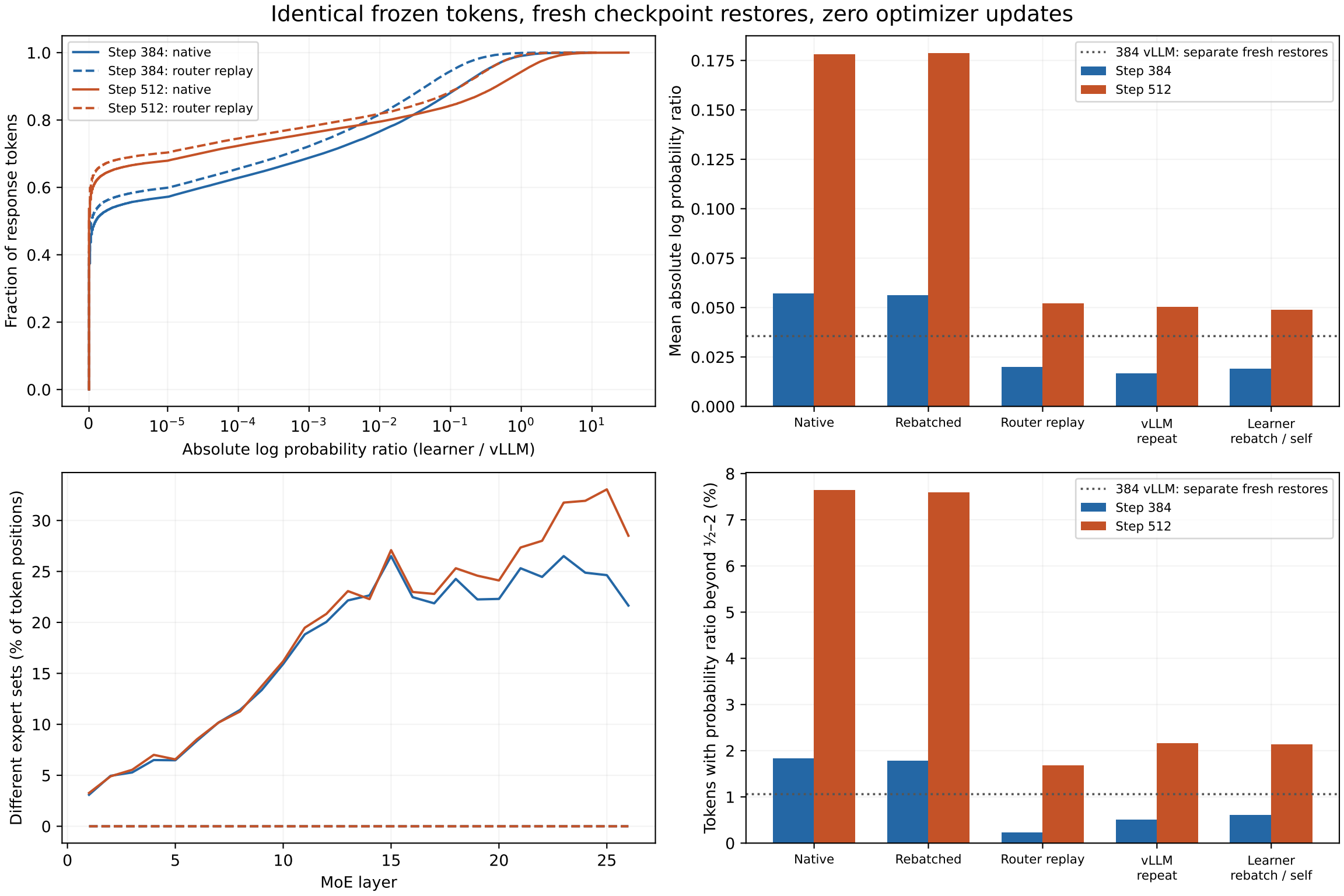
An auxiliary decode control generates eight responses of at most 256 tokens to each of four questions (G01, G03, P03, S09). It compares probabilities recorded during autoregressive decoding against vLLM prefill rescoring and the three learner modes. These 32 short responses differ from the common fixed-token panel. A shared request seed is used, so repeated prompts are not independent samples. The completed auxiliary control's coverage is listed separately from the primary comparison.

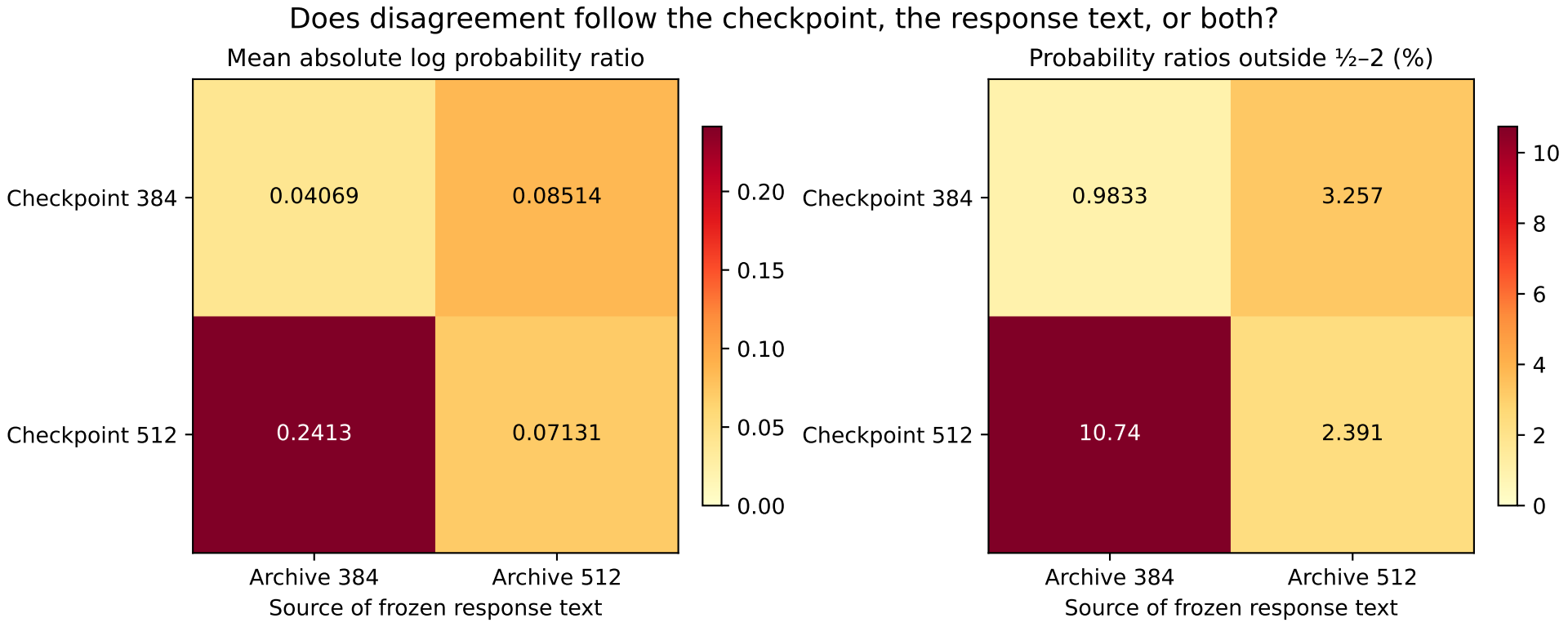
Definitions: $\delta = \log p_{learner}(\text{token} \mid \text{prefix}) - \log p_{vLLM}(\text{token} \mid \text{prefix})$, using natural logs. Mean $|\delta|$ measures the typical absolute log-probability difference. $k3 = \exp(\delta) - 1 - \delta$, averaged over scored tokens; on this fixed panel it is a descriptive mismatch statistic, not an unbiased current-policy KL estimate. Beyond 2× means $|\delta| > \log 2$, so the probability ratio is below 1/2 or above 2. $\text{ESS/N} = (\sum r)^2 / (N \sum r^2)$, where $r = \exp(\delta)$; a few large ratios lower it. Expert-set disagreement counts input positions/layers where the unordered top-four expert IDs differ.

Frozen-panel SHA-256: f2ccda6b0e073d8c67a0f8bcfcf788ace40ae5aa7c3f2b4d59a06dd2ed520241

MarinSkyRL commit: 5663f6129f53bb6aaa09549fa6cb423571503e94. vLLM build: marin.39e62869693c.cu132; PyTorch 2.13.0+cu132. Primary scoring definitions are identical between the two runs; only the auxiliary request changed for API compatibility. The accompanying evidence-manifest.json records archive locations, checkpoint identities, source hashes and verification artifacts.

The following two pages contain the fixed-token distributions and checkpoint-by-text comparison.


Identical frozen tokens, fresh checkpoint restores, zero optimizer updates
Step 384: native
Step 384: router replay
Step 512: native
Step 512: router replay
Fraction of response tokens
Absolute log probability ratio (learner / vLLM)
384 vLLM: separate fresh restores
Step 384
Step 512
Mean absolute log probability ratio
Native
Rebatched
Router replay
vLLM repeat
Learner rebatch / self
Different expert sets (% of token positions)
MoE layer
Tokens with probability ratio beyond ½–2 (%)


Does disagreement follow the checkpoint, the response text, or both?
Mean absolute log probability ratio
Checkpoint 384
Checkpoint 512
0.04069
0.08514
0.2413
0.07131
0.20
0.15
0.10
0.05
0.00
Archive 384
Archive 512
Source of frozen response text
Probability ratios outside ½–2 (%)
Checkpoint 384
Checkpoint 512
0.9833
3.257
10.74
2.391
10
8
6
4
2
0
Archive 384
Archive 512
Source of frozen response text